

PEPERTECT

Admin Panel

Product Requirements Document

Version 1.0 | June 2025

Indian Stock Market Paper Trading Platform

Prepared for: Pepertect Development Team

Classification: Confidential

Status: Draft for Review

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1. Document Overview

Purpose

This document defines all features, requirements, and specifications for the Pepertect Admin Panel — a separate web application for administrators to manage the Pepertect paper trading platform.

Scope

- Separate admin panel at `admin.pepertect.vercel.app`
- Connected to same backend (API routes + Database) as `pepertect.vercel.app`
- Real-time sync via Socket.io — admin actions reflect instantly on frontend
- Role-based access control (Super Admin, Admin, Moderator)

Technology Stack

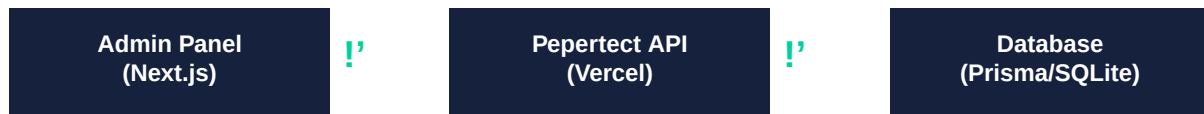
Technology	Details
Framework	Next.js 16 with App Router + TypeScript
UI Library	shadcn/ui (New York style) + Tailwind CSS 4
State	Zustand (client) + TanStack Query (server)
Real-time	Socket.io Client !' <code>pepertect-ws.onrender.com</code>
Icons	Lucide React
Charts	Recharts (bar, line, pie, area)

Key URLs

- Admin Panel: `admin.pepertect.vercel.app`
- Main App: `pepertect.vercel.app`
- WebSocket: `pepertect-ws.onrender.com`
- API Base: `pepertect.vercel.app/api/admin/*`

2. System Architecture

Architecture Diagram



Real-time Flow



Authentication Flow

- Admin Login !' Email + Password !' POST /api/admin/auth/login
- Server validates credentials !' Returns JWT with admin role claim
- Client stores JWT in httpOnly cookie + Zustand
- All /api/admin/* routes verify JWT + admin role
- Token auto-refreshes 5 minutes before expiry

Data Flow Principles

- Admin panel NEVER directly accesses the database
- All data operations go through Pepertect API routes
- Admin API routes have verifyAdmin() middleware
- Shared database: Same SQLite via Prisma ORM
- Real-time: Socket.io connects to same Render WS service

3. Authentication & Authorization

Admin Login Requirements

- Email + Password authentication (NO Google OAuth for admin)
- Failed login: Show generic error, increment attempt counter
- Lock account after 5 failed attempts (30 min cooldown)
- Remember me: 30 day session vs 24hr default

Role System

Role	Capabilities
Super Admin	Full access: Users, Settings, Other Admins, Delete, Impersonate
Admin	Users, Trades, Notifications, Market Data, Content
Moderator	Read-only + Users view/ban + Notifications send

JWT Token Structure

```
{ "sub": "admin_id", "role": "super_admin", "email": "admin@pepertext.com", "iat": 1718000000, "exp": 1718086400 }
```

API Middleware: verifyAdmin()

- Check Authorization header for Bearer token
- Decode JWT ! Verify signature ! Check role claim
- Reject if token expired or role not in [super_admin, admin, moderator]
- Log all admin API calls to audit trail

Password Requirements

- Minimum 8 characters
- At least 1 uppercase letter
- At least 1 number
- At least 1 special character
- Cannot reuse last 3 passwords

Session Management

- JWT expiry: 24 hours (configurable)
- Auto-refresh: 5 minutes before expiry
- Single session per admin (new login invalidates old token)
- Logout: Clear cookie + Blacklist token for remaining duration

Future: Two-factor authentication (TOTP) for Super Admin role

4. Dashboard Requirements

Stats Cards (Top Row)

Card	Data Source	
Total Users	COUNT users	Number + growth
Active Traders Today	COUNT trades (today)	Number + trend
Total Trades Today	COUNT trades (today)	Number + avg price
Platform Volume	SUM trade value (today)	1 formatted + bar chart

Charts Section

- User Growth: Line chart — Daily/Weekly/Monthly toggle
- Trade Volume: Bar chart — Daily/Weekly/Monthly toggle
- P&L Distribution: Pie chart — Profit/Loss/Neutral users
- Sector Distribution: Donut chart — Where users are invested

System Health Panel

Indicator	Source	
Upstox WS	GET /api/admin/market/status	Ø=βâ Connected
Upstox Token	GET /api/upstox/token-status	' Valid / & p Exp
Server Uptime	Render service	HH:MM:SS form
API Response Time	Middleware tracking	Avg ms + trend
WS Connections	Render health endpoint	Count + last 1hr

Quick Actions

- Send Broadcast !' Opens notification compose modal
- Toggle Maintenance !' Confirmation dialog !' Socket.io emit to all users
- Refresh Market Cache !' POST /api/admin/market/cache/clear
- Refresh Upstox Token !' POST /api/upstox/refresh-token

Recent Activity Feed

- Latest 10 admin actions (who did what, when)
- Latest 10 user registrations (name, email, time)
- Latest 10 trades (user, stock, type, amount)
- Auto-refresh every 30 seconds via TanStack Query

5. Users Management

Users List Page

Primary admin view for managing all registered users on the platform.

Table Columns

Column	Details
Avatar	32px circular user avatar or initials
Name	Full name (clickable !' detail page)
Email	Email address (verified/unverified badge)
Phone	Phone number (if provided)
Role	User / Admin badge
Status	⊘=β Active / ⊘=Ÿ4 Banned / &« Inactive
Joined	Date (relative: "2 days ago")
Last Active	Relative timestamp
Trades	Total trade count
P&L	Total P&L with color (green/red)
Actions	View Edit More %¼ (Ban/Delete/Reset)

Search & Filters

- Search: By name, email, phone (debounced 300ms)
- Filter by Role: User, Admin, All
- Filter by Status: Active, Banned, Inactive, All
- Filter by Date: Joined after/before, Last active after/before
- Sort: Name (A-Z), Join date (newest/oldest), Last active, Trade count, P&L
- Pagination: 25 / 50 / 100 per page

Bulk Actions

- Select multiple users via checkboxes
- Bulk Ban — requires reason text
- Bulk Delete — requires typing "DELETE" to confirm
- Bulk Export — CSV download with all user data

User Detail Page

Comprehensive view of individual user with all their data.

- Profile Section: Name, email, phone, avatar, role, status, join date, Google linked
- Trading Summary: Total trades, Win rate %, Total P&L, Current portfolio value, Cash balance
- Balance History: Line chart — virtual cash changes over time
- Recent Trades: Last 20 trades in mini-table
- Current Holdings: Open positions mini-table
- Activity Log: Last 20 actions (login, trades, settings changes)
- Admin Notes: Add/view notes (only visible to admins)

User Actions

Action	Confirmation
Edit Profile	Save button
Reset Password	Confirm dialog
Ban User	Reason required + duration
Unban User	Confirm dialog

None (admin-on

Email sent to us

User session inv

User can login a

Delete User

Type "DELETE" to confirm

Promote Role

Confirm dialog

Reset Balance

Confirm dialog

Impersonate

Super Admin only

API Endpoints

GET /api/

admin/

users —

List with pagination, filters, search

GET /api/

admin/

users/[id]

— User

detail +

trading

summary

PUT /api/

admin/

users/[id]

—

Update

profile

data

DELETE

/api/

admin/

users/[id]

— Delete

user permanently

POST /

api/admin/

users/[id]/

ban —

Ban with

reason +

duration

POST /

api/admin/

users/[id]/

unban —

Unban

user

POST /

api/admin/

users/[id]/

reset-

password

—

Generate

temp

password

POST /

api/admin/

users/[id]/

reset-

balance

— Reset

to

1,00,000

POST /

api/admin/

users/[id]/

impersonate

— Get

read-only

user token

GET /api/

admin/

users/

export —

CSV

download

6. Trades & Orders

Trades List Page

Column	Details
User	Name (clickable !' user detail)
Stock	Symbol + name (clickable !' stock detail)
Type	BUY (green) / SELL (red) badge
Quantity	Number of shares/lots
Price	Execution price per unit
Total Value	Qty × Price (¹ formatted)
P&L	Realized P&L for sell trades (green/red)
Status	Filled / Pending / Cancelled
Time	Exact timestamp (IST)

Filters

- By User: Search by name/email
- By Stock: Search by symbol/name
- By Type: Buy, Sell, All
- By Status: Filled, Pending, Cancelled, All
- By Date Range: From – To date picker
- Sort: Time (newest/oldest), Price, Quantity, P&L
- Export: CSV download of filtered trades

Open Positions View

Aggregate view of all currently held positions across all users.

- Stock-wise aggregation: Total qty, Avg price, Current value, Combined P&L
- User-wise breakdown: Drill-down per stock to see individual holders
- Risk indicators: Positions with >50% loss highlighted red, >30% warning yellow
- Sector tags: Banking, IT, Pharma, Auto, etc.

Order Management

- View order details: User, Stock, Type, Qty, Price, Status, Timestamp
- Force Cancel: Admin can cancel any pending order
- View fill history: Partial fills, timestamps
- Order statistics: Fill rate %, Average execution time

Trade Analytics

- Most Traded Stocks: Top 10 by volume and frequency
- Active Trading Hours: Heatmap (hour × day of week)
- Average Trade Size: By stock, by user segment
- Buy vs Sell Ratio: Overall and per stock
- Popular Option Strikes: For NIFTY/BANKNIFTY options
- Futures Analysis: Open interest, Lot size distribution

API Endpoints

GET /api/
admin/
trades —
List with
filters, pag
ination,
export
GET /api/

admin/
positions
— All
open
positions
with aggre
gation
GET /api/
admin/
orders —
All orders
with
status
filter
DELETE
/api/
admin/
orders/[id]
— Force
cancel an
order
GET /api/
admin/
trades/
analytics
— Trade
analytics
data
GET /api/
admin/
trades/
export —
CSV
download

7. Portfolio & P&L

Portfolio Overview

Column	Details
User	Name (clickable !' user detail)
Portfolio Value	Holdings + Cash balance (¹ formatted)
Cash Balance	Available virtual cash
Invested Amount	Total invested in positions
P&L	Unrealized + Realized (green/red)
P&L %	Percentage return
Trades	Total trade count

Aggregate Statistics

- Total Platform P&L: Sum of all user P&L
- Average P&L per User: Mean return across all users
- P&L Distribution: Pie chart — Profit / Loss / Neutral user count
- Investment Distribution: Bar chart — How much invested vs cash

Top Performers

- Top 10 Gainers: By absolute P&L and by P&L %
- Top 10 Losers: By absolute P&L and by P&L %
- Most Active: By trade count
- Most Consistent: Highest win rate (min 10 trades)

Sector Analysis

- Sector-wise investment: Where is money concentrated
- Sector-wise P&L: Which sectors are profitable
- Most popular sectors: By number of investors
- Concentration risk: If >40% in one sector !' warning

Balance Tracking

- Total virtual cash in system: Sum of all user balances
- Total invested: Sum of all position values
- Cash vs Invested ratio chart
- Daily balance change tracker

API Endpoints

GET /api/
admin/
portfolios
— All
portfolios
with P&L

GET /api/
admin/
portfolios/
analytics
— Aggreg
ate stats

GET /api/
admin/
portfolios/
top-
gainers
— Top 10
gainers
GET /api/

admin/
portfolios/
top-losers
— Top
10 losers
GET /api/
admin/
portfolios/
sectors
— Sector
analysis

8. Market Data Management

Upstox Connection Status

Field	Display
Connection	Ø=βâ Connected / Ø=Ý4 Disconnected
Data Source	upstox-ws-v3 (0.1-0.2s) / rest-polling (1s)
Access Token	Valid / Expiring soon / Expired
Token Obtained	Timestamp of last token grant
Token Expiry	Calculated expiry time
Last Refresh	When token was last refreshed
Feed URL	Last authorized WS URL (truncated)

WebSocket Monitoring

- Connection status with reconnect count
- Subscribed instruments: 5 indices + 80+ stocks
- Messages received per minute (throughput)
- Uptime since last connection
- Last data received timestamp
- Latency estimate based on tick frequency

API Usage Monitoring

Metric	Details
Total API Calls Today	Count + hourly breakdown chart
Rate Limit Status	Remaining / Limit per window
Average Response Time	ms + p95 + p99
Error Rate	Percentage + error type breakdown
Fallback Activations	How many times Dhan/Yahoo was used instead of Upstox

Cache Management

- View all cached data keys with TTL and last update time
- Clear specific cache: Market Live, Option Chain, Stock Quotes
- Clear all cache (nuclear option)
- Force refresh: Fetch fresh data from Upstox immediately

Stock Management

- Active stocks list: View all 80+ stocks with instrument keys
- Add new stock: Enter symbol + instrument key
- Remove stock: Remove from active trading list
- Enable/Disable: Temporarily disable specific stocks
- Bulk import: CSV upload of stocks

API Endpoints

GET /api/
 admin/
 market/
 status —
 Upstox +
 WS
 status
 overview
POST /
 api/admin/

market/
cache/
clear —
Clear
cache
(specific
or all)
GET /api/
admin/
market/
api-usage
— API
call stats
PUT /api/
admin/
market/
stocks —
Manage
active
stock list
POST /
api/admin/
market/
refresh
— Force
data
refresh

9. Notifications

Broadcast Notifications

Send real-time notifications to all users or specific users from admin panel.

Create Notification Form

Field	Details
Title	Max 100 chars, required
Message	Max 500 chars, rich text editor
Type	Info (blue) / Warning (yellow) / Success (green) / Error (red)
Target	All Users / Specific User / User Segment
Schedule	Send Now / Schedule for future time
Priority	Normal / High (shows as banner)

Preview & Send Flow

- Step 1: Fill form !' Preview notification as user would see it
- Step 2: Confirm !' POST /api/admin/notifications/broadcast
- Step 3: Socket.io event "admin:notification" emitted to all clients
- Step 4: Track delivery: Sent count, Read count, Dismissed count

Announcement Banners

- Create banner: Text, Type (info/warning/urgent), Position (top/bottom)
- Dismissible: Can user close it? Yes/No
- Active Period: Start time !' End time
- Multiple banners: Stacked, newest on top
- Socket.io event: "admin:banner" !' Frontend shows banner in real-time

Notification History

- Table: Title, Type, Target, Sent At, Read Count, Actions
- Filter by: Type, Date range, Target type
- Resend: Re-send same notification
- Delivery stats: Sent / Read / Dismissed percentages

Real-time Events (Socket.io)

Event	Payload
admin:notification	{ title, message, type }
admin:banner	{ text, type, dismissible }
admin:maintenance	{ enabled, message }
admin:user-banned	{ userId }

Toast notification

Banner shows o

Maintenance ov

User session inv

API Endpoints

POST /
api/admin/
notifications/
broadcast
— Send
to all/
segment
POST /
api/admin/
notifications/
individual
— Send

to
specific
user
GET /api/
admin/
notifications/
history
— Notifica
tion log
POST /
api/admin/
banners
— Create
announce
ment
banner
PUT /api/
admin/
banners/
[id] —
Update
banner
DELETE
/api/
admin/
banners/
[id] —
Remove
banner

10. Platform Settings

Upstox Configuration

- API Key: Displayed masked (sk_****xxx), view full on click
- API Secret: Always masked, never shown in full
- Access Token: Status + expiry time + refresh button
- Auth Callback URL: Display current, editable
- Manual OAuth Trigger: Button '!' redirects to Upstox auth
- Token Refresh: Manual + Auto-refresh status

Trading Configuration

Setting	Default	
Default Virtual Cash	₹1,00,000	Starting balance
Market Open Time	09:15 IST	When trading begins
Market Close Time	15:30 IST	When trading ends
Trading Days	Mon-Fri	Configurable per week
Max Position Size	Unlimited	Max qty per stock
Max Daily Trades	Unlimited	Max trades per day
Lot Size (F&O)	As per exchange	Override lot size

Feature Flags

Feature	Default	
Equity Trading	ON	Buy/sell stocks
F&O Trading	ON	Futures & Options
Options Trading	ON	Option chain + t
Short Selling	OFF	Sell without hold
After-hours Orders	OFF	Place orders wh
Watchlist	ON	Custom watchlis
Paper Trading Mode	ON	If OFF, platform

Maintenance Mode

- Toggle: Enable/Disable with one click
- Custom Message: "Platform under maintenance, back in 30 min"
- Estimated Duration: Display expected downtime
- Scheduled: Set future time for auto-enable/disable
- Real-time Effect: Socket.io "admin:maintenance" !' All users see overlay
- Admin Access: Admins can still access platform during maintenance

Rate Limits

Limit	Default	
API Calls / Minute	60	Per user rate lim
WS Messages / Second	10	Per client WebS
Login Attempts / Hour	5	After which acco
Trade Orders / Minute	10	Prevent spam tr

API Endpoints

GET /api/

admin/
settings
— All
platform
settings
PUT /api/
admin/
settings
—
Update
settings
(bulk)
GET /api/
admin/
settings/
upstox
— Upstox-
specific
config
POST /
api/admin/
settings/
upstox/
refresh-
token —
Refresh
Upstox
token
PUT /api/
admin/
settings/
feature-
flags —
Toggle
features
POST /
api/admin/
settings/
maintenance
—
Toggle ma
intenance
mode

11. Security & Audit

Admin Action Log

Immutable audit trail of all admin actions on the platform.

Field	Details
Timestamp	Exact time of action (IST)
Admin	Who performed the action
Action	What was done (user.ban, settings.update, etc.)
Target	What was affected (user ID, setting key)
Details	Before/After values for changes
IP Address	Where the action was performed from

Admin action logs CANNOT be deleted or modified. They are the immutable audit trail.

Suspicious Activity Detection

- Multiple accounts from same IP: Flag if >2 accounts share IP
- Abnormally high trade volume: >100 trades/hour per user
- Rapid login/logout: >10 sessions in 1 hour
- Flagged accounts list: Review + take action
- Auto-flag threshold: Configurable via settings

Security Dashboard

- Active sessions count (admin + user)
- Failed login attempts (last 24h)
- Rate limit violations (last 24h)
- Blocked IPs list
- Account recovery requests

API Endpoints

GET /api/
admin/
audit/logs

—
Action log
with filters

GET /api/
admin/
audit/
suspicious

—
Flagged
accounts

GET /api/
admin/
audit/
sessions

— Active
sessions

POST /
api/admin/
audit/
block-ip

— Block
an IP
address

12. Content Management

FAQ Management

- List all FAQs with drag-and-drop reordering
- Add new FAQ: Question + Answer (rich text)
- Edit existing FAQ
- Delete FAQ (with confirmation)
- Categories: General, Trading, Account, Technical

Help & Support Content

- Help articles: CRUD with rich text editor
- Categories: Getting Started, Trading Guide, Account Management
- Article preview: See how it looks before publishing
- SEO fields: Meta title, description for each article

Learning Section

- Tutorial management: Add/edit/delete tutorials
- Content types: Text, Video embed, Interactive
- Difficulty levels: Beginner, Intermediate, Advanced
- Publish/draft workflow

Legal Pages

- Terms of Service: Edit content
- Privacy Policy: Edit content
- Disclaimer: Edit content (important for trading platform)
- Version tracking: See previous versions

API Endpoints

GET /api/

admin/
content/

faq —

List all

FAQs

POST /

api/admin/
content/

faq —

Create

FAQ

PUT /api/

admin/
content/

faq/[id]

—

Update

FAQ

DELETE

/api/

admin/
content/

faq/[id]

— Delete

FAQ

GET /api/

admin/
content/

articles

— List

help

articles

PUT /api/

admin/

content/

articles/
[id] —
Update
article

13. Non-Functional Requirements

Performance

- Dashboard initial load: < 2 seconds
- Table pages (users, trades): < 1 second
- Search results: < 500ms
- Real-time updates: < 200ms from admin action to user visible
- CSV export: < 5 seconds for 10,000 records

Security

- HTTPS only (enforced)
- CSRF protection on all state-changing endpoints
- XSS prevention (sanitized output)
- SQL injection prevention (Prisma parameterized queries)
- Content Security Policy headers
- Secure cookies (httpOnly, secure, sameSite=strict)

Availability & Scalability

- Target: 99.9% uptime
- Support 10,000+ registered users
- Support 100+ concurrent admin sessions
- Database: Handle 100,000+ trade records
- WebSocket: Handle 1,000+ simultaneous connections

Responsive Design

- Desktop: 1440px+ (primary target)
- Tablet: 768px+ (secondary)
- Mobile: Not supported (admin-only, desktop-first)

Accessibility

- WCAG 2.1 Level AA compliance
- Keyboard navigation for all actions
- Screen reader compatible
- Color contrast ratio "e 4.5:1

Browser Support

- Google Chrome: Latest 2 versions (primary)
- Firefox: Latest 2 versions
- Safari: Latest 2 versions
- Edge: Latest 2 versions

14. API Security Requirements

Authentication

- All /api/admin/* routes require valid JWT with admin role
- Token in Authorization: Bearer header
- Token verification on every request (no caching of auth)
- Expired tokens !' 401 Unauthorized response
- Non-admin tokens !' 403 Forbidden response

Rate Limiting

- 100 requests per minute per admin IP
- Burst limit: 20 requests per second
- 429 Too Many Requests when exceeded
- Rate limit headers: X-RateLimit-Remaining, X-RateLimit-Reset

Input Validation

- All inputs validated with Zod schemas
- String length limits enforced
- No raw HTML accepted (XSS prevention)
- File uploads: Max 5MB, allowed types only

CORS Policy

- Only allow: admin.pepertect.vercel.app
- Block all other origins
- Credentials: allowed
- Methods: GET, POST, PUT, DELETE only

Audit Logging

- Log every API request: method, path, admin, IP, timestamp
- Log all data modifications: before/after values
- Log all failed auth attempts
- Logs stored in database (immutable)
- Log retention: 90 days minimum

Error Handling

- No stack traces in production responses
- No database details in error messages
- Generic error messages for 500 errors
- Specific errors only for validation (400) and auth (401/403)

15. Implementation Priority

P0 — Must Have (Week 1)

- Admin Authentication: Login, JWT, verifyAdmin middleware
- Dashboard: Stats cards, System health, Quick actions
- Users CRUD: List, View, Edit, Ban/Unban, Delete
- Trades View: List with filters, Open positions
- Admin API Routes: Core /api/admin/* endpoints

P1 — Should Have (Week 2)

- Portfolio & P&L: Overview, Top performers, Sector analysis
- Notifications: Broadcast, Individual, History
- Market Data: Upstox status, WS monitoring, Cache management
- Platform Settings: Trading config, Feature flags, Rate limits
- Real-time Events: Socket.io integration for admin actions

P2 — Nice to Have (Week 3)

- Content Management: FAQ, Help articles, Legal pages
- Advanced Analytics: Charts, Heatmaps, Trade patterns
- Security Audit: Action logs, Suspicious activity, IP tracking
- Announcement Banners: Create/manage/dismiss
- Bulk Operations: Multi-select, Bulk ban/delete/export

P3 — Future Enhancements

- User Impersonation: Read-only session as user
- Two-Factor Authentication: TOTP for Super Admin
- Custom Dashboards: Drag-and-drop widget layout
- API Playground: Test API endpoints from admin panel
- Webhook Management: Configure outgoing webhooks
- Automated Reports: Daily/weekly email reports
- AI-powered Insights: Anomaly detection, user behavior analysis

Total estimated development time: 3-4 weeks for P0-P2 features. P3 features to be planned based on user feedback and platform growth.